



Customer: DRINKWATER PRODUCTS LLC

|          |        |
|----------|--------|
| P.O. No. | 017822 |
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| S.O. No. | 525751 |
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## Material Test Report

Heat Code: ET6

Ameriforge Ltd

13770 Industrial Rd. Houston, TX 77015

Sales: (713) 393-4200

ISO 9001:2015 Certified

|                                                                                        |       |                                                            |                     |         |
|----------------------------------------------------------------------------------------|-------|------------------------------------------------------------|---------------------|---------|
| NOV-R&M ENERGY SYS<br>500 CONROE PARK WEST DR<br>CONROE, TX 77303                      | 15278 | PO: 4033411                                                | Sales Order: 191006 | Line: 2 |
|                                                                                        |       | Item Code: 9999H60002-0031R                                | Qty Shipped: 799    |         |
|                                                                                        |       | Item Desc: 02.00 150-900CAP A350LF2 17518043-001 NRM CDFRG |                     |         |
|                                                                                        |       | Mill Heat: C217479                                         |                     |         |
| Spec: ASME SA105N/SA350 (17) ASME SA105N/SA350 (17) LF2 CL1 & NOV 61-0072 rev 1 (Caps) |       |                                                            |                     |         |

ET6

| Element       | (%wt) | Ladle | Product | EPCRA | CAS#      | Element       | (%wt) | Ladle | Product | EPCRA | CAS#      |
|---------------|-------|-------|---------|-------|-----------|---------------|-------|-------|---------|-------|-----------|
| C Carbon      | 0.19  | ✓     |         |       |           | Cr Chromium   | 0.11  | ✓     |         |       |           |
| Mn Manganese  | 1.15  | ✓     |         | Y     | 7439-96-5 | Mo Molybdenum | 0.02  | ✓     |         | Y     | 7440-47-3 |
| P Phosphorous | 0.008 | ✓     |         |       |           | V Vanadium    | 0.035 | ✓     |         |       |           |
| S Sulphur     | 0.01  | ✓     |         |       |           | Cb Columbium  | 0.00  | ✓     |         |       |           |
| Si Silicon    | 0.27  | ✓     |         |       |           | Al Aluminum   | 0.02  | ✓     |         |       |           |
| Cu Copper     | 0.15  | ✓     |         | Y     | 7440-50-8 | C.E.          | 0.43  | ✓     |         |       |           |
| Ni Nickel     | 0.06  | ✓     |         | Y     | 7440-02-0 | CuNiCrMoV     | 0.38  |       |         |       |           |

| Mechanical Testing              |                    | Heat Treat         |                |
|---------------------------------|--------------------|--------------------|----------------|
| Test Lab                        | AFG                | Norm. Temp. (F)    | 1675           |
| Test Bar Size                   | SACRIFICIAL PIECE  | Norm. (Hrs)        | 2.00           |
| HBW                             | 143 - 149          | Norm. Cooling Type | AIR            |
| Elg (%)                         | 28.8               |                    |                |
| RA (%)                          | 64.4               | Other              |                |
| Tensile Specimen Size (in)      | .352               | Supplier           | STEEL DYNAMICS |
| Tensile (ksi)                   | 77.0               | Country of Melt    | USA            |
| Yield (ksi)                     | 49.4               | Country of Forging | USA            |
| Gauge Length (in)               | 1.50               | Vacuum Degassed    | Y              |
| Charpy Temp (F)                 | -50                |                    |                |
| Charpy (ft lbs)                 | 46 - 40 - 49       |                    |                |
| Avg Energy 3 Specimens (ft-lbs) | 45                 |                    |                |
| Min Energy 1 Specimens (ft-lbs) | 40                 |                    |                |
| % Shear                         | 20 - 20 - 20       |                    |                |
| Lat. Exp. (in)                  | .047 - .038 - .051 |                    |                |

by ALP

Date 2/1/23

cap

The recording of false, fictitious, or fraudulent statement or entries on this document may be punishable as a felony under federal statute.

EPCRA Supplier Notification: This product may contain one or more toxic chemicals subject to the reporting requirements of Section 313 of the Emergency Planning and Community Right-to-Know Act (Title III of the Superfund Amendments and Reauthorization Act of 1986) and 40 C.F.R. Part 372. Potentially reportable chemicals are indicated with a checkmark in the "EPCRA" column and a Chemical Abstract Services (CAS) registry number is provided for each such chemical in addition to the percent by weight of the chemical present in this product. It is your responsibility alone to determine whether your facility is required to submit a Toxic Release Inventory Report under EPCRA Section 313.

Certification No.: 1622151

Certification Date: 1/23/2023

Issued By: Michelle Gutierrez

Page 1 of 2

This report is issued in compliance with the requirements of EN10204 3.1 / ISO 10474 3.1.b

# Material Test Report

Heat Code: ET6

Ameriforge Ltd

13770 Industrial Rd. Houston, TX 77015

ISO 9001:2015 Certified

Sales: (713) 393-4200

|                                                                                        |       |                                                            |                     |         |
|----------------------------------------------------------------------------------------|-------|------------------------------------------------------------|---------------------|---------|
| NOV-R&M ENERGY SYS                                                                     | 15278 | PO: 4033411                                                | Sales Order: 191006 | Line: 2 |
| 500 CONROE PARK WEST DR                                                                |       | Item Code: 9999H60002-0031R                                | Qty Shipped: 799    |         |
| CONROE, TX 77303                                                                       |       | Item Desc: 02.00 150-900CAP A350LF2 17518043-001 NRM CDFRG |                     |         |
|                                                                                        |       | Mill Heat: C217479                                         |                     |         |
| Spec: ASME SA105N/SA350 (17) ASME SA105N/SA350 (17) LF2 CL1 & NOV 61-0072 rev 1 (Caps) |       |                                                            |                     |         |

Product compliant with NACE MR0175/ISO 15156-2:2015, Annex A (A.2.1.2, A2.1.3).  
Product compliant with NACE MR0103/ISO 17945:2015, Section 13.1.1 and 13.1.3.

- \* No Weld repair performed.
- \* Ladle Chemical Analysis results are reported from the raw material suppliers MTR.
- \* Tensile taken from a round specimen. Measurement of elongation after fracture.
- \* Yield strength was determined using the .2% offset method, unless specified.
- \* All material supplied under this order is certified to be free of mercury contamination and no mercury bearing equipment was used in manufacturing, fabrication or testing.
- \* When reported on MTR: Tensile Testing Per ASTM E8; Brinell Hardness Per ASTM E10; Charpy Testing Per ASTM E23
- \* AF Global manufactures materials according to a certified Quality Management System conforming to ISO 9001:2015 and PED certified to 2014/68/EU, Annex 1, Section 4.3.
- Unless otherwise noted on PO, Dimensions of all standard pipe flanges and flange fittings (NPS 1/2 to NPS 24) per ASME B16.5-2020, large diameter steel flanges (NPS 26 to 60) per ASME B16.47-2020, Orifice flanges per ASME B16.36-2020.
- \* AFGlobal uses the latest edition of ASTM E29 (Standard practice for using significant digits in test data to determine conformance with specifications), to claim conformance or nonconformance to a specification. When a accept/reject determination is required AFGlobal uses a test uncertainty ratio (TUR) of 1:1 or higher as acceptable.
- \* DNV PED Certificate No 10000425664-PA-ACCREDIA-USA Valid thru December 03, 2023.

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Certification No.: 1622151

Certification Date: 1/23/2023

Issued By: Michelle Gutierrez

  
Dusty Bannon - Quality Analyst

This report is issued in compliance with the requirements of EN10204 3.1 / ISO 10474 3.1.b

## Material Test Report

Heat Code: ED6

Ameriforge Ltd

13770 Industrial Rd. Houston, TX 77015

Sales: (713) 393-4200

ISO 9001:2015 Certified

|                                                                   |       |                                                             |                     |         |
|-------------------------------------------------------------------|-------|-------------------------------------------------------------|---------------------|---------|
| NOV-R&M ENERGY SYS<br>500 CONROE PARK WEST DR<br>CONROE, TX 77303 | 15278 | PO: 4029723                                                 | Sales Order: 188181 | Line: 2 |
|                                                                   |       | Item Code: 9999H61002-0031R                                 | Qty Shipped: 1,612  |         |
|                                                                   |       | Item Desc: 2.00 150-900 HUB A350LF2 -17518204-001 NRM CDFRG |                     |         |
|                                                                   |       | Mill Heat: C216029                                          |                     |         |
| Spec: ASME SA350/SA350M(21) LF2 CL1 & NOV 61-0072 Rev 01 (Hubs)   |       |                                                             |                     |         |

| Element       | (%wt) | Ladle | Product | EPCRA | CAS#      | Element       | (%wt) | Ladle | Product | EPCRA | CAS#      |
|---------------|-------|-------|---------|-------|-----------|---------------|-------|-------|---------|-------|-----------|
| C Carbon      | 0.19  | ✓     | 0.20    |       |           | Cr Chromium   | 0.07  | ✓     | 0.07    | Y     | 7440-47-3 |
| Mn Manganese  | 1.12  | ✓     | 1.15    | Y     | 7439-96-5 | Mo Molybdenum | 0.02  | ✓     | 0.01    |       |           |
| P Phosphorous | 0.009 | ✓     | 0.008   |       |           | V Vanadium    | 0.03  | ✓     | 0.03    |       |           |
| S Sulphur     | 0.01  | ✓     | 0.01    |       |           | Cb Columbium  | 0.00  | ✓     | 0.01    |       |           |
| Si Silicon    | 0.25  | ✓     | 0.24    |       |           | Al Aluminum   | 0.02  | ✓     | 0.02    |       |           |
| Cu Copper     | 0.15  | ✓     | 0.14    | Y     | 7440-50-8 | C.E.          | 0.42  | ✓     |         |       |           |
| Ni Nickel     | 0.06  | ✓     | 0.05    | Y     | 7440-02-0 |               |       |       |         |       |           |

| Mechanical Testing         |                   | Heat Treat                  |                |
|----------------------------|-------------------|-----------------------------|----------------|
| Test Lab                   | AFG               | Norm. Temp. (F)             | 1675           |
| Test Bar Size              | SACRIFICIAL PIECE | Norm. (Hrs)                 | 4              |
| HBW                        | 149 - 149         | Norm. Cooling Type          | AIR            |
| ELg (%)                    | 34                |                             |                |
| RA (%)                     | 63                |                             |                |
| Tensile Specimen Size (in) | .245              | Supplier                    | STEEL DYNAMICS |
| Tensile (ksi)              | 82.4              | Vacuum Degassed             | Y              |
| Yield (ksi)                | 52.3              | Gr.Sz                       | 8              |
| Gauge Length (in)          | 1                 | Fully Killed                | Y              |
| Specimen Orientation       | LONGITUDINAL      | Made to Fine Grain Practice | YES            |
| LCVN Charpy Temp. (F)      | -50               |                             |                |
| LCVN (ft lbs)              | 30 - 60 - 44      |                             |                |
| LAvg 3(ft-lbs)             | 44                |                             |                |
| 1 Min (ft-lbs)             | 30                |                             |                |
| 1/2 Shear                  | 10 - 40 - 30      |                             |                |
| Lat. Exp.                  | .028 - .054 - 041 |                             |                |

Approved for Manufacture

by [Signature]

Date 11/1/22

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Certification No.: 1610135

Certification Date: 10/10/2022

Issued By: Michelle Simien

This report is issued in compliance with the requirements of EN10204 3.1 / ISO 10474 3.1.b

# Material Test Report

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|                                                                   |       |                                                             |                     |                    |
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|                                                                   |       | Item Desc: 2.00 150-900 HUB A350LF2 -17518204-001 NRM CDFRG |                     |                    |
|                                                                   |       | Mill Heat: C216029                                          |                     |                    |
| Spec: ASME SA350/SA350M(21) LF2 CL1 & NOV 61-0072 Rev 01 (Hubs)   |       |                                                             |                     |                    |

Product compliant with NACE MR0175/ISO 15156-2:2015, Annex A (A.2.1.2, A2.1.3).  
Product compliant with NACE MR0103/ISO 17945:2015, Section 13.1.1 and 13.1.3.  
TAG: H61002

- \* No Weld repair performed.
- \* Ladle Chemical Analysis results are reported from the raw material suppliers MTR.
- \* Tensile taken from a round specimen. Measurement of elongation after fracture.
- \* Yield strength was determined using the .2% offset method, unless specified.
- \* All material supplied under this order is certified to be free of mercury contamination and no mercury bearing equipment was used in manufacturing, fabrication or testing.
- \* When reported on MTR: Tensile Testing Per ASTM E8; Brinell Hardness Per ASTM E10; Charpy Testing Per ASTM E23
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Certification Date: 10/10/2022  
Issued By: Michelle Simien

  
Dusty Bannon - Quality Analyst

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